

videos. Further, the language of image captions is somewhat restricted in terms of grammatical structure and lexical items, making the analysis of long-tail phenomena or highly abstract language challenging (Ferraro et al., 2015; Alikhani and Stone, 2019). Future work could use our framework to explore other meaning representations, such as symbolic models or videos (though doing so involves overcoming further dataset and modeling challenges). Still, the language-neutral nature and rich information content of images allows us to study groundedness for a wide range of words, languages, and linguistic contexts.

Noting that a model’s surprisal is negative log probability, we can view groundedness as a *difference in surprisal*, corresponding to how much more expected the token is under the grounded model than under the textual model. As such, the PMI should rarely take on negative values—because the captioning model has more information (both image and text) than the language model (text only). However, some tokens, such as those that are highly grammatical or structural, should be close to 0.

In this work, we study the groundedness of *word classes*. Drawing inspiration from functionalist typology, we treat a word class $\mathcal{C}_i$ as a label selected by a linguist for a word in its context. We make an assumption that this label is independent of our meaning representation given a word’s context, allowing us to define the following joint distribution:

$$p(\mathcal{C}_i, m \mid \mathbf{w}_{<t}) = \sum_{w_t \in \mathcal{W}} [p(\mathcal{C}_i \mid w_t, \mathbf{w}_{<t}) p(w_t, m \mid \mathbf{w}_{<t})]. \quad (2)$$

We can then formulate the mutual information between a word class and meaning as the expected value of the PMI between each token labeled with that class, and the token’s associated image:

$$I[\mathcal{C}_i; m \mid \mathbf{w}_{<t}] = \mathbb{E}_{p(\mathcal{C}_i, m \mid \mathbf{w}_{<t})} \left[\log \frac{p(w_t \mid \mathbf{w}_{<t}, m)}{p(w_t \mid \mathbf{w}_{<t})} \right]. \quad (3)$$

Given our factorization of the joint, we can perform a Monte Carlo estimation of the expectation by simply averaging groundedness over all the tokens tagged with $\mathcal{C}_i$ in the data $\mathcal{D}$:

$$\hat{I}[\mathcal{C}_i; m \mid \mathbf{w}_{<t}] = \sum_{(m, \mathbf{w}_{<t}) \in \mathcal{D}} \frac{\mathbb{1}_{\mathcal{C}_{w_t}=\mathcal{C}_i} \log \frac{p_\phi(w_t \mid \mathbf{w}_{<t}, m)}{p_\theta(w_t \mid \mathbf{w}_{<t})}}{\sum_{w_t \in \mathcal{D}} \mathbb{1}_{\mathcal{C}_{w_t}=\mathcal{C}_i}} \quad (4)$$

Model	Gemma PT	PaliGemma CT	COCO-35L FT
Img. Cap. LM	A	 A	 A
	A	 A	A

Table 1: We match the data points on which the language model and image captioning model were trained. The three datasets are the Gemma pre-training mixture (PT), PaliGemma multimodal data for continued training (CT), and COCO image-caption pairs for fine-tuning (FT). Symbols indicate whether models are trained on text data (**A**) or on multimodal data ( **A**).

where $\mathbb{1}_{\mathcal{C}_{w_t}=\mathcal{C}_i}$ is 1 when a token’s class is $\mathcal{C}_i$ and 0 otherwise. We note that our groundedness measure and our mutual information estimates are conditional on *linguistic context*. As such, words which are very grounded in one context could be hardly grounded in another, due to disambiguating information in the preceding context. Some information about m will be generally conveyed by $\mathbf{w}_{<t}$; however, our mutual information estimates are aggregated over all contexts in which a word class occurs, and on average this contribution is small.

4 Experimental setup

Captioning model $p_\phi(w_t \mid \mathbf{w}_{<t}, m)$ As our image captioning model, we use the recently released PaliGemma model (Beyer et al., 2024). This model is by far the state-of-the-art among publicly available multilingual image captioning models. PaliGemma consists of an image encoder, initialized from the SigLIP-So400m model (Zhai et al., 2023), and a transformer decoder language model, initialized from the Gemma-2B language model (Gemma, 2024). A linear projection maps from the image encoder space to a sequence of 256 tokens in the language model’s embedding space. The whole system is then trained on a mix of vision-and-language datasets, including the unreleased WebLI dataset with 10 billion image-caption pairs in 109 languages (Chen et al., 2023), and the CC3M-35L dataset consisting of 3 million image-caption pairs in each of 35 languages (Thapliyal et al., 2022).

While PaliGemma is a general-purpose vision-and-language model, it is designed to be fine-tuned on and applied to individual tasks. As such, we use the open-source paligemma-3b-ft-coco35-224 checkpoint for multilingual captioning, which has been fine-tuned on COCO-35L.

Language model $p_\theta(w_t \mid \mathbf{w}_{<t})$ Our aim is to use a language model as similar to our captioning

model $p_\phi(w_t|\mathbf{w}_{<t}, m)$ as possible. This is critical to getting good (P)MI estimates, which relies on estimating a difference in surprisal between the two models. If the language model is not adapted to the image captioning domain, it may under-estimate the probability of particular words, leading to an over-estimation of mutual information. We therefore aim to *match* the training data between the language model and image captioning model, such that they see the same set of captions.

To do so, we initialize our language model with the weights from the pretrained PaliGemma model paligemma-3b-pt-224. However, out of the box, the decoder behaves degenerately when no image is provided, so we need to adapt the model to not expect image information and to match the training data of the captioning model. To do so, we fine-tune the language model on the *captions only* from the COCO-35L dataset. In this way, we ensure the models have observed the same data during training and are adapted to the same domain, and are thus maximally comparable. Table 1 summarizes the data matching between the two models. Further implementational and POS tagging details are in Appendix A.

Evaluation Datasets We also need multilingual image captioning datasets for evaluation which are not observed during training. For this, we measure groundedness on three separate datasets, each with its own strengths and weaknesses. First, we use **Crossmodal-3600**. This dataset includes captions for 3,600 images across a range of cultures, manually captioned by fluent speakers of 36 typologically diverse languages. However, it is relatively small per language compared to other datasets. Further, the independence of the captions means that there is greater diversity in what aspects of an image are being described across languages (Liu et al., 2021; Ye et al., 2024; Berger and Ponti, 2024).

Our second dataset, the validation set of **COCO-35L**, addresses several of these issues. It is larger, with 5 captions each for 5000 images and 35 languages,² yielding 25,000 captions per language. Further, the captions are machine translations of each other, ensuring more comparable semantic content across languages (Beekhuizen et al., 2017) at the expense of centering the perspective of English speakers and machine translation issues.

Finally, we consider **Multi30K**. This dataset

²Crossmodal-3600 and COCO-35L cover the same languages with the exception of Quechua.

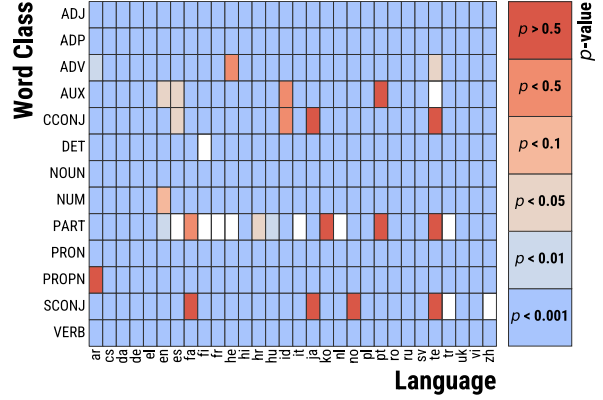


Figure 2: Heatmap of mutual information estimates across parts of speech in thirty languages. Cells show the statistical significance of a word class’s groundedness ($MI > 0$). Unattested classes are white. Some functional classes display non-significant levels of groundedness in several languages, while lexical classes dominantly show highly significant grounding.

comprises 30,000 images captioned 5 times each in English, with a single caption per image manually translated into French, German, Czech, and Arabic. This dataset is therefore large on the individual language level, but with limited language coverage. It has the comparability of being translated and the trustworthiness of human translation, but may still be vulnerable to translationese. By looking at all three of these datasets for similar generalizations about the relationship between groundedness and part of speech, we obtain a picture that is robust to the weaknesses of the individual datasets.

5 Results

The following sections quantitatively investigate the trends in our groundedness measure across languages and word classes. We begin by examining which word classes exhibit significant groundedness (Section 5.1), followed by an analysis of cross-linguistic trends and their consistency (5.2 and 5.3). Finally, we relate our findings to contentfulness-related psycholinguistic norms (5.4).

5.1 Which word classes are grounded?

We first investigate the evidence for groundedness in each word class—that is, for each part of speech, we ask whether its estimated mutual information with the image is significantly greater than zero.

To compute significance levels, we use a one-sample permutation test. Taking the set of PMIs for a part of speech (POS) in a language, we sample up to 500 PMIs at a time from all datasets and

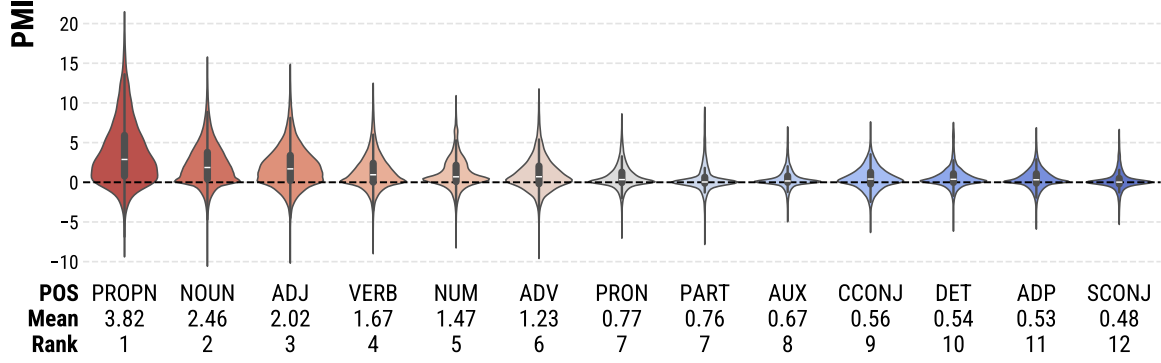


Figure 3: Word token level distributions of the groundedness measure (PMI) across all languages and datasets, grouped by part of speech (word class). We also report the estimated marginal mean and ranking of each word class. Colors are based on the ranking of classes, rather than their average PMIs. Overall, the distribution and estimated ranking of word classes strongly suggest our groundedness measure quantitatively captures the distinction between lexical and functional classes.

randomly permute their signs (assign + or - with equal probability to each PMI value), then average these values to produce a new estimate of mutual information (MI). We repeat this process to produce 10^5 permuted estimates. By measuring how often our estimate based on the observed data is greater than the permuted estimate, we obtain the p -value,³ i.e., the probability that our observations would have occurred under the null hypothesis of $MI = 0$.

Results are shown in Figure 2. Overall, the results suggest most or all word classes contribute some information about the image they describe—in line with theories in linguistics that emphasize the lexical aspects of categories which are traditionally considered functional (Corver and Riemsdijk, 2001; Bisang, 2017). Interestingly, subordinating and coordinating conjunctions do not consistently reject the null hypothesis, suggesting there is little evidence the image is informative for how many clauses a speaker uses to describe an image.

5.2 Which word classes are more grounded?

We hypothesize that the cross-linguistically consistent trends in word class groundedness correspond to a cline which is a continuous analogue of the lexical–functional word class distinction. To isolate the contribution of word class identity to mutual information cross-linguistically, we compute estimated marginal means (EMMs) for each word class’s groundedness,⁴ and perform a post-hoc pairwise comparison test of the means.⁵ The results of this analysis are displayed in Figure 3. All pair-

wise comparisons except between pronouns and particles are statistically significant, leading to a near total ranking of word classes. We find that lexical word classes (Proper nouns, nouns, adjectives, verbs, numbers, and adverbs) have higher groundedness than functional word classes (particles, auxiliaries, conjunctions, determiners, and adpositions), with pronouns ranking together with particles at the upper end of the functional categories. The ranking corroborates ideas from cognitive linguistics which place nouns, adjectives, and verbs along a lexical–functional continuum, with nouns > adjectives > verbs (Rauhut, 2023). On the other hand, it does not neatly align with ideas in linguistic theory about adpositions as a semi-lexical class (Corver and Riemsdijk, 2001), which suggest they should behave more like other lexical classes compared to functional classes. Instead we see similar or greater mutual information for other functional classes, suggesting they could be more meaning-bearing than traditionally viewed.

5.3 How consistent is word class groundedness across languages?

We quantify the strength of the association between groundedness and word class on two levels: language-level MI estimates (Figure 1), and token-level PMI (Figure 3). The first level quantifies how consistent languages are in the groundedness of word classes, while the second level quantifies how much word class drives the groundedness of individual tokens. In both cases, we use ANOVA to estimate the amount of the variance in groundedness explained by word class.

³We use the Benjamini and Yekutieli (2001) corrections.

⁴Averaged over values of language and dataset.

⁵Using Šidák corrections; significance threshold = 0.01.